

Sample Surveys

Class 5

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Fall 2026

Previously
on
Sample Surveys...

Recap

- Confidence intervals (CI) for $\bar{Y}$.
 - Exact CI **cannot** be computed.
 - Rely on **asymptotic approximations**.
 - Under some conditions, and for large $N, n, N - n$, the sampling distribution of $\bar{y}$ can be approximated by a **Normal** distribution.
 - For SRS, approximate 95% CI for $\bar{Y}$ is of the form

$$[\bar{y} - 1.96 \times \widehat{SE}(\bar{y}), \bar{y} + 1.96 \times \widehat{SE}(\bar{y})]$$

- Sample size determination.
 - **Margin of error**: e (absolute) and r (relative).
 - Connect n with the margin of error via some equation. $\rightarrow$ **equate with CI for $\bar{Y}$** .
 - For SRSWR, $n_{WR} = \frac{z_{\alpha/2}^2 \sigma^2}{e^2}$.
 - For SRSWOR, $n_{WOR} = \frac{z_{\alpha/2}^2 S^2}{e^2 + \frac{z_{\alpha/2}^2}{N}} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$.
 - Desired choice depends on **unknown** population quantities.

Choosing n for estimating population proportions

- Parameter of interest: P .

- $$n_{WR} = \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2}, \quad n_{WOR} \approx \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{e^2 + \frac{z_{\frac{\alpha}{2}}^2 P(1-P)}{N}}.$$

- Worst-case approach:** When the population is the **most unpredictable**.
 - Assume the state of **maximum uncertainty**, which requires the **largest sample size** to cut through the noise.
 - Corresponds to $P = 0.5$. Get n for this choice of P .
 - If true P is 0.5, we will have surveyed **exactly enough** people.
 - If true P is something else (e.g., 0.20), our n will be **larger than strictly necessary**.

Worst-case approach: an example

- What proportion of the population uses net banking on their mobile phones?
- Set margin of error $e = 0.05$.
- $$n_{WR} = \frac{1.96^2 \times 0.25}{0.05^2} = 384.16.$$
 Choose n to be 385.
- If we make e **higher**, e.g., $e = 0.1$ resulting n will be **lower**, i.e., $n = 97$.
- In general, what if we focus on **relative** margin of error r ?

Determining n for SRS: general guidelines

- With absolute error, the resulting n depends on S^2 (or σ^2).
- How to plug in?
 - Bound by the worst-case value (e.g., as in the population proportion case).
 - **Pilot survey:** Do a **small survey before the actual survey** to get initial estimates of these parameters.
 - Use previous studies or data available in the literature.
 - If nothing else is available, **guess the variance**. Use a theoretical model if needed.
- With relative error, the resulting n depends on the **CV** of Y in the population.
- How to plug in?
 - Similar ideas as before.
 - Interestingly, CV tends to be **more stable** over time and is **easier to guess** in advance. (e.g., house prices)
- We often have **multiple study variables**.
 - Different variables demand different sample sizes.
 - Focus on one or two important variables (a compromise).

Choice of n : a hypothesis testing perspective

- Previous approach: estimation perspective.
 - Trap the true parameter within a margin of error with high probability.
 - Choice of n is driven by e (or r) and α (confidence level).

Hypothesis testing approach

- Goal: To make a binary, actionable **decision** about the unknown parameter.
 - E.g., Is the true average rent in the city strictly greater than ₹6,000?
 - To test $H_0 : \bar{Y} = 6000$ vs $H_1 : \bar{Y} > 6000$.
 - Choice of n is driven by two distinct types of **decision making risks**.
 - $\alpha = P(\text{Type-I error})$. $\rightarrow$ The risk of a **false positive**.
 - $\beta = P(\text{Type-II error})$. $\rightarrow$ The risk of a **false negative**.
- Power of the test: $1 - \beta$.
- Focus on simple null vs simple alternative.

$$H_0 : \bar{Y} = \mu_0 \quad \text{vs} \quad H_1 : \bar{Y} = \mu_1 (> \mu_0).$$

*If the true mean is actually μ_1 , I want to be **confident with probability $1 - \beta$** that my test will catch it. **What sample size do I need?***

Choice of n : a hypothesis testing perspective

- Testing problem: $H_0 : \bar{Y} = \mu_0$ vs $H_1 : \bar{Y} = \mu_1 (> \mu_0)$.
- **Effect size:** $\Delta = \mu_1 - \mu_0$. $\rightarrow$ Helps to choose the relevant μ_1 .
 - **Minimum detectable effect:** How big does the shift need to be before we actually care enough to do something about it?
 - House rent example: Maybe we only care if the rent has jumped by at least ₹500. Choose $\Delta = 500$.

Example: Worker strike

- The workers' union in a food delivery company is thinking about going on strike to demand a higher salary due to a potential increase in fuel expense.
- **Baseline:** Historical data shows the average daily fuel expense is ₹150 ($= \mu_0$).
- **Tipping point:** A ₹50 increase wipes out the exact payout of one average food delivery. So, $\Delta = 50$, hence $\mu_1 = 200$.

Example: Manufacturing and quality control ($\mu_1 < \mu_0$ case)

- A factory produces steel bolts for bridges. The baseline tensile strength (μ_0) is 500 megapascals (MPa). They suspect the new batch of steel may be weaker.
- If the strength drops to 490 MPa, the bridge becomes unsafe. So, $\mu_1 = 490$.

Choice of n for testing: A general recipe

- Set your effect size (and hence the alternative hypothesis).
- Set the values of the probability of Type-I error and Power. E.g., $\alpha = 0.05$, $1 - \beta = 0.8$.
- Create an equation connecting n to the above quantities and solve for n .
- Check if sampling with this choice of n is feasible given the budget. If not, **revise your criterion**.
 - Increase α : we increase the chance of accidentally claiming a shift happened **when it actually hasn't**. (rarely done though)
 - Accept a higher risk of **missing the shift**: Increase β , i.e., lower the Power.
 - Accept only caring about **larger, more obvious** changes: Increase Δ .
- $H_0 : \bar{Y} = \mu_0$ vs $H_1 : \bar{Y} = \mu_1 (> \mu_0)$
- Consider a right-tailed test. **How to perform the test?**
- **Asymptotic approximations**: $\frac{\bar{y} - \bar{Y}}{SE(\bar{y})} \overset{\text{approx}}{\sim} \mathcal{N}(0, 1)$.
- **Pretend** for now that you know $SE(\bar{y})$.
- Test statistic: $t = \frac{\bar{y} - \mu_0}{SE(\bar{y})}$. Reject H_0 at level α if $t > z_\alpha$.
 - z_α is the $(1 - \alpha)$ th quantile of a standard Normal distribution.

SRS: finding the right n for testing

- Compute the **Power of the test**, and solve for n .
- For SRSWR, $n_{WR} = \frac{(z_{\alpha} + z_{\beta})^2 \sigma^2}{\Delta^2}$.
- For SRSWOR, $n_{WOR} = \frac{(z_{\alpha} + z_{\beta})^2 S^2}{\Delta^2 + \frac{(z_{\alpha} + z_{\beta})^2 S^2}{N}}$. So, $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$. (same as before)
- For two sided tests, $n_{WR} \approx \frac{(z_{\frac{\alpha}{2}} + z_{\beta})^2 \sigma^2}{\Delta^2}$, $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$.
- **Plug in σ^2** : Similar idea as before.

Example: Diabetes case study

- An investigator hypothesizes that in people free of diabetes, fasting blood glucose is **higher** in those who drink at least 2 cups of coffee per day.
- Historical data: mean fasting blood glucose level in people free of diabetes is reported as 95.0 mg/dL with an sd of 9.8 mg/dL.
- If the mean blood glucose level in people who drink at least 2 cups of coffee per day is 100 mg/dL, this would be important clinically.
- How many patients should be enrolled in the study (under SRSWR) to ensure that the power of a two-sided test at 5% level is 80% to detect this difference?
- $n_{WR} = 30.185$. Set $n = 31$.

Estimating population proportions: choice of n

- $H_0 : P = P_0$ vs $H_1 : P = P_1 (> P_0)$.
- Consider a right-tailed test.
- For SRSWR, $n_{WR} = \frac{(z_\alpha + z_\beta)^2 P_0(1-P_0)}{\Delta^2}$. $\rightarrow$ everything is known.
- For SRSWOR, $n_{WOR} \approx \frac{n_{WR}}{1 + \frac{n_{WR}}{N}}$. (same as before)
- Two-sided tests: similar approximations as before.

Example: Heart disease case study

- A recent report from the Framingham Heart Study indicated that 26% of people free of heart disease had elevated LDL cholesterol levels, defined as LDL > 159 mg/dL.⁹
- An investigator hypothesizes that, among patients with a history of heart disease, a higher proportion will have elevated LDL cholesterol.
- How many patients should be studied to ensure that the power of a two-sided test at 5% level is 90% to detect a 0.05 difference in the proportion with elevated LDL cholesterol?
- $n_{WR} = 808.75$. Set $n = 809$.